



National Institute of Technology Rourkela
Department of Computer Science Engineering
Introduction to AI-ML (CS2011)
Mid Semester Examination, B.Tech 2nd year

Duration: 2 hrs

Autumn 2024, 3rd Semester

Total marks: 30

Attempt all questions

1. Answer the following questions. [2 × 4 = 8]
 - (a) Given the mappings f, ϕ , with any input $x \in R^d$, represent through the given functions, the difference between classification and regression. Assume a binary classification with classes $\{-, +\}$.
 - (b) Differentiate between supervised and unsupervised learning with examples? Identify whether the following mapping is supervised or not. Also determine the number of classes: $f : R^d \rightarrow \{+, *, -\}$ [R^d represents ' d ' dimensional space.]
 - (c) What is difference between accuracy metrics: F1 score, ROC and AUC? Explain with diagrams wherever necessary.
 - (d) Express the confusion matrix in terms of its components. What do you understand by sensitivity and specificity? Write the formula for each.
2. (a) Define AI with four use cases. [2+3 = 5]
 - (b) For each of the following tasks, give a PEAS description of the task environment:
 - Driverless car
 - Virtual personal assistants
 - Satellite image analysis system
3. (a) The 'Missionary and Cannibals' problem has a famous history in AI. Here is one statement of the problem: Three missionaries and three Cannibals come to a river. There is a boat on their side of the river that can be used by either one or two people. How should they use this boat to cross the river in such a way that Cannibals never outnumber Missionaries on either side of the river?
Specify the form of state descriptions, the starting state and the goal state for this problem. Draw the entire state description graph and label its nodes. (You need to include only those states that are 'legal' — that is, states in which Cannibals do not outnumber missionaries on either side of the river.) [3]

- (b) Imagine there is a robot in room 'A' (initial state), and it needs to go to room 'Z' (goal state) as given in the following Figure 1 . Draw the state space tree for the above scenario by the following approach: [2+2 = 4]

- Depth-first search
- Breadth-first search

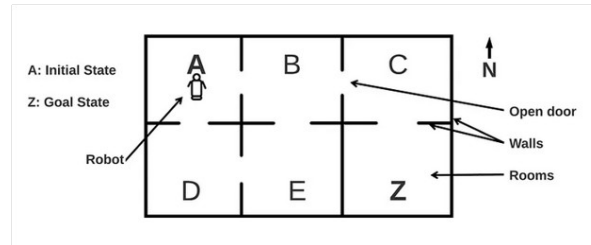


Figure 1: State space diagram

4. (a) What is K-fold cross validation (CV)? Given a data set containing 100 samples with input/output pairs (x_i, y_i) , answer the following: What is the value of K with 10 samples/fold? [2]
- (b) A machine learning model is used to classify whether emails are spam or not. The following confusion matrix summarises the classification results: Compute the accuracy, precision, recall, and F1-Score. Which one is the best measure between Recall and Precision? Justify your answer. [2+1=3]

	Predicted: Spam	Predicted: Not spam
Actual: Spam	80	20
Actual: Not spam	10	90

5. (a) Write short notes on Bayes' classification model explicitly stating the formulations. Assume that, the two classes are w_1 (accept) and w_2 (reject) with the input sample being x . [3]
- (b) Consider a dataset with the following prior probabilities:
 $P(Class1) = 0.4$; $P(Class2) = 0.6$
 $P(X = 1|Class1) = 0.8$; $P(X = 1|Class2) = 0.5$
 Find $P(Class1|X = 1)$ and $P(Class2|X = 1)$? [2]